

Project Design Phase

Solution Architecture

Date	26 June 2026
Team ID	SWTID-2026-8141
Project Name	Heart Disease Analysis and Visualization Dashboard
Maximum Marks	4 Marks

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best technology solution to solve the existing heart-disease data-fragmentation problem.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications for making decisions during implementation.

Architecture Overview (Layered View):

Layer	Description
Data Layer	Raw heart disease datasets: HeartDiseaseData_Clean.csv, HeartDiseaseIndia.csv, hospital_cardiac_center_list.csv, HighRiskPatients-HeartDiseaseDatabase.csv
Processing Layer	Python/Pandas cleaning and preprocessing; MySQL database storage; SQL queries for analysis
Visualization Layer	Tableau Desktop worksheets, interactive dashboard, and story; published to Tableau Public
Application Layer	Flask web application (Python) serving HTML/CSS/JavaScript pages that embed Tableau visualizations
Presentation Layer	Responsive web interface accessed by clinicians, researchers, hospital administrators, and patients

Data Flow: Heart Disease Datasets → Data Cleaning & Preprocessing → MySQL Database → SQL Analysis → Tableau Desktop → Interactive Dashboard & Story → Tableau Public → Flask Web Application → End User.